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Demographics as Correlates to Performance among Ambulance Personnel Providing Out-of-Hospital Cardiac Arrest (OHCA) Management: Basis for a Proposed Personnel Competency Program

Marydell Claire B. Dado, Kenneth C.Eupena, Keisha Danielle T. Tolentino, Dr. Pricila B. Marzan, Dr. Lucy L. Teves *(ORCID No. 0000-0003-0939-2824),* Vincent Louis P. Duncano

ABSTRACT - *This research assessed the relationship between the demographics and performance among ambulance personnel providing Out-of-Hospital Cardiac Arrest (OHCA) management leading to the development of a personnel competency program. Descriptive correlational research method was employed involving 75 ambulance personnel from 11 ambulance providers servicing Surigaodel Norte. Descriptive statistics was used to describe the profile and their performance in OHCA Management. Likewise, inferential statistics specifically Pearson r correlation was used to determine the relationship between variables. The study shows significant relationship existing between years of experience in ERS and ambulance personnel performance in transporting and managing OHCA patients.*

Keywords: *nursing education, out-of-hospital cardiac arrest, ambulance personnel, patient transport and management, OHCA management, Philippines*

INTRODUCTION

Medical emergencies can happen at any time and can be fatal, particularly during pandemic times when everyone is on the lookout for ways to help in an emergency. It could occur at any moment and in any place. To achieve a positive outcome, one must always be prepared to cope with emergencies, such as cardiac arrest. It frequently leads in death if not addressed within minutes. Outside of a hospital and away from competent medical treatment, the majority of cardiac arrests occur. With each passing minute, your odds of survival dwindle considerably.

Sudden cardiac arrest is responsible for 40-50 percent of all cardiovascular deaths worldwide. Ischemic heart disease, which is described as a disease characterized by a reduced blood supply to the heart and is linked to cardiac arrest, was the leading cause of death in 2018, according to the Philippine Statistics Authority.

In 2012, hypertension was one of the top causes of morbidity in the Caraga Region, with 11,627 instances per 100,000 population (Department of Health, 2012). 256 cardiovascular attacks/hypertension were reported in Surigao City's 2016 Annual Report (47 percent male and 53 percent female). Medical transport services were primarily supplied by the City Emergency Response Services (ERS) Unit to the Caraga Regional Hospital and private hospitals. Medical transport is also available for patients who require advanced care in the adjacent cities of Butuan, Cagayan de Oro, and Davao City (Gesta, 2016).

Out-of-hospital cardiac arrest patients should have a better chance of surviving if emergency providers are available. In industrialized countries like the United States and Australia, there have been numerous advancements in the care of out-of-hospital cardiac arrest patients, but this is not the situation in third-world countries (M. Daya et al, 2015). The availability of emergency providers is critical in the Philippines, since the medical area is frequently disregarded

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and undeveloped. Though there are other factors that contribute to a better patient outcome, such as the time and location of the arrest and bystander CPR, ambulance personnel's out-of-hospital cardiac arrest management is given the highest priority.

Following cardiac arrest, cardiopulmonary resuscitation (CPR) and defibrillation are the first line of therapy and the basis for saving lives. For treating out-of-hospital cardiac arrest patients, emergency response services have long been the only option. It has been demonstrated that failure to achieve the prescribed standard of treatment reduces the likelihood of patient survival in out-of-hospital cardiac arrest (Dyson, 2017).

It has been an observation, Emergency room nurses and doctors at Caraga Regional Hospitals have repeatedly tried to resuscitate individuals who appear to have had cardiac arrest at home, and in the majority of cases, ER health care providers have lost the battle. There is a need for information on rescuer resuscitation competency in out-of-hospital cardiac arrest to improve patient survival. The readiness of the rescuers has a significant impact on the outcomes of their care. It's critical to consider not only the readiness of the staff, but also the resources they'll require to accomplish their tasks properly.

Conceptual Framework of the Study

The study is based on concepts from Nola J. Pender's Health Promotion Model, revised in 1996, and the World Health Organization's Pre-Hospital Treatment Protocol of Adult Cardiac Arrest. According to the HPM paradigm, each person has distinct personal characteristics and experiences that influence their later actions. The set of variables for behaviorally specific knowledge and affect has a lot of motivational power. Nursing interventions can change these variables. The desired behavioral consequence and the HPM's end aim is health-promoting behavior.

The WHO protocol establishes a standard for pre-hospital cardiac arrest management. Participants' age, sex, civil status, greatest educational attainment, years of experience linked to ERS, and trainings attended relevant to ERS make up their profile.

Scene arrival and size-up; evaluation priorities; ambulance stretcher operation; treatment (high-quality CPR and AED); medicine use and storage; exception protocol; and transfer decision are all part of normal patient care for pre-hospital cardiac arrest.

The interplay of these variables is depicted in the study's schematic diagram. The independent variable is the profile of the participants in terms of their age, sex, civil status, greatest educational attainment, years of experience in ERS, trainings attended linked to ERS, and ambulance facilities, which is contained in the box on the left. The dependent variables are the usual patient care of adult cardiac arrest in the box on the right

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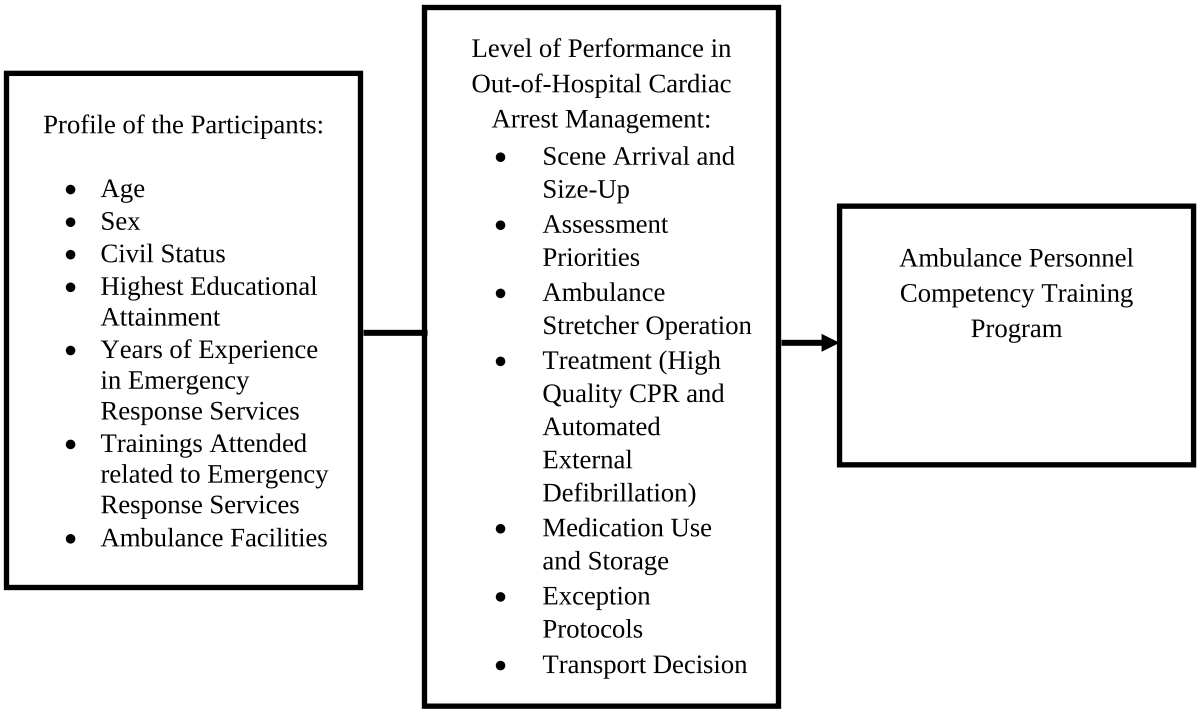

Figure 1. Schematic Diagram of the Study

Statement of the Problem

The main problem of the study is to assess the level performance of ambulance personnel in the management of out-of-hospital cardiac arrest.

Specifically, it sought to answer the following questions:

1. What is the profile of ambulance personnel in terms of:
   - 1.1 Age;
   - 1.2 Sex;
   - 1.3 Civil Status;
   - 1.4 Highest Educational Attainment;
   - 1.5 Years of Experience in Emergency Response Services (ERS); and
   - 1.6 Trainings attended related to ERS
   - 1.7 Ambulance Facilities
2. What is the level of performance of the ambulance personnel in terms of:
   - 2.1 Scene Arrival and Size-up
   - 2.2 Assessment Priorities
   - 2.3 Ambulance Stretcher Operation
   - 2.4 Treatment (High Quality CPR and Defibrillation)
   - 2.5 Medication Use and Storage
   - 2.6 Exception Protocols
   - 2.7 Transport Decision
3. Is there a significant relationship between the respondents profile and level of performance?
4. Based on the findings, what program may be proposed?

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Hypothesis

There is no significant relationship between the respondent's profile and level of performance?

METHOD

The study used a descriptive-survey design to describe the performance of ambulance workers in Surigaodel Norte in transporting and managing OHCA patients. The participants were identified using a convenient sampling procedure.

Participants

The study's participants are the 32 emergency room nurses from the four (4) hospitals, 30 Rural Health nurses from the LGU emergency response services, 3 midwives from the Rural Health Unit and 10 ambulance drivers from the LGU emergency response services in Surigao del Norte, which are ambulance providers.

The ambulance providers are the 4 hospitals in Surigao del Norte: Caraga Regional Hospital, St. Paul Surigao University Hospital, St. Paul Surigao University Hospital, Surigao Del Norte Provincial Hospital and the Emergency Response Services of the 7 LGUs namely, Surigao City, Tagana-an, San Francisco, Malimono, Tubod, Sison, Mainit. The distribution of participants is shown below.

Table 1. Distribution of the Participants from the Ambulance Providers

	Ambulance Providers	Sample Size
Hospitals	Caraga Regional Hospital	8
	St. Paul Surigao University Hospital	5
	Surigao Medical Center	7
	Surigao Del Norte Provincial Hospital	12
Emergency Response Services	Surigao City	13
	Tagana-an	5
	San Francisco	5
	Malimono	5
	Tubod	5
	Sison	5
	Mainit	5
	TOTAL	75

Instruments

The data was gathered using a survey questionnaire based on the World Health Organization's Pre-Hospital Treatment Protocol of Adult Cardiac Arrest. The participants' age, sex, civil status, greatest educational attainment, years of experience connected to ERS, and trainings attended relevant to ERS are all listed in the first section. The assessment of their degree of performance in regard to out-of-hospital cardiac arrest care is the second phase. The Department of Health's guidelines for a properly equipped ambulance were used to create the ambulance checklist. It has all of an ambulance's supplies, equipment, and medications.

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The data gathered were analyzed based on the following parameters

Scale	Parameter	Verbal Interpretation	Qualitative Description
4	3.25-4.00	Strongly Agree (SA)	Exemplary (E)
3	2.50-3.24	Agree (A)	Proficient (P)
2	1.75-2.49	Disagree (D)	Advanced Beginner (AB)
1	1.00-1.74	Strongly Disagree (SD)	Novice (N)

Data Analysis

The following are the statistical tools that were used in this study:

Percentage and Frequency Count. Used to treat the participants' profiles.

Mean and Standard Deviation. Were used to determine the participants' degree of performance in respect to out-of-hospital cardiac arrest care.

Pearson Correlation. This tool was used to determine the relationship between the profile of the participants and their level of performance.

RESULTS AND DISCUSSION

This chapter presents the results and discussions of the study according to the order of the problems presented in chapter 1.

Table 2. Profile of the Participants

Variables	f (*n* = 75)	Percent (%)
Age		
20-24 years old	7	9.33
25-29 years old	17	22.67
30-34 years old	32	42.67
35-39 years old	11	14.67
40-44 years old	5	6.67
45 years old and above	3	4.00
Sex		
Male	27	36.00
Female	48	64.00
Civil Status		
Single	33	44.00
Married	42	56.00
Highest Educational Attainment		
BS in Nursing	57	76.00
Midwifery	3	4.00
Non allied health course	10	13.33
BSN with Masters Degree	5	6.67
Years of Experience in ERS		
Less than a year	7	9.33
1-3 years	37	49.33

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4-6 years	18	24.00
7-10 years	7	9.33
10 years and above	6	8.00
Trainings Attended related to ERS		
Standard First Aid	66	88.00
Basic Life Support	76	100.00
Advanced Cardiac Life Support	41	54.67
Hazards Communication and Respiratory Safety	9	12.00
Out-of-Hospital Cardiac Arrest Management	8	10.67
Emergency Medical Technician Training	9	12.00

According to the table, the bulk of the participants are between the ages of 30 and 34, with a frequency of 32, accounting for 42.47 percent of the total number of participants. It is followed by the age group of 25 to 29 years old, which has a frequency of 17 and accounts for 22.67 percent of all participants. Looking at the other variables, it's clear that young adults (19 to 40 years old) make up the majority of the population. Individuals need to build personal, loving relationships with other people at this stage of Erik Erikson's Psychosocial Theory. They look at relationships that lead to longer-term commitments with people who aren't family members. Finding work that inspires passion and happiness is one approach to accomplish this. Failure leads to loneliness and isolation, whilst success leads to close relationships. As people get older, the number of people working in this profession decreases rapidly.

There are 48 female participants out of 75 overall, accounting for 64% of the total number of participants, and 27 male participants, accounting for 36% of the total number of participants. The majority of the participants are clearly female. The findings contradict the perception that men are more dominant in this field than women. Women have an easier time connecting with patients. As a result, creates a better working environment and makes it easier to carry out interventions (Satterwhite, 2013).

In terms of civil status, 42 participants are married, accounting for 56 percent of the total number of participants, while 33 are single, accounting for 44 percent of the total number of participants. Within a partnership, marriage fosters a sense of commitment, safety, and caring. It has a good effect on people's psychological stress (Chin, Murphy, Janicki-Deverts, & Cohen, 2017). Ambulance staff can do their responsibilities more efficiently when they are less stressed, which results in a better patient outcome.

With a frequency of 57, or 76 % of participants, the bulk of the participants had a Bachelor of Science in Nursing as their highest educational attainment. It is followed by Non-Allied Health Course, which has a frequency of 10 and accounts for 13.33 % participants, Masters Degree 5, which stands for 6.67 %, and Midwifery 3, which represents for 4% of the total participants.

The research participants in hospitals were prioritized as emergency nurses. Because their workers are only trained for patient transport and only have the driver for any emergency, there is a severe dearth of participants in the field of emergency response in towns. As a result, nurses from rural health units were chosen as research participants since they are well-versed in cardiac arrest and its treatment. Patients with severe illnesses are occasionally accompanied by community health nurses, especially during long-distance transport.

When it came to years of experience in ERS, the majority of the participants (44.7% of the total number of participants) had 1-3 years of experience. Then comes 4-6 years of experience, which accounts for 25.88% of the total number of participants. Work exposure for ambulance personnel allows them to practice and improve their skills. As a result, one of the factors that leads to patient survival is exposure to cardiac arrest situations. A study discovered that improving patient survival takes 1 to 4 years (L. H. Soo et al, 1998), which is evident in the study's findings.

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There are multiple responses to their ERS trainings. The Basic Life Support had a frequency of 75, indicating that all participants had been trained in it. It was followed by Standard First Aid, which had a frequency of 65, indicating that 86.67 percent of the participants had received training in it. The two trainings described are the most popular among participants, implying that they are lacking in critical training for out-of-hospital cardiac arrest management.

Ambulance Facilities

Table 3. The Facilities of Ambulance Providers

Ambulance Providers	*f (n=88)*	%
Caraga Regional Hospital	80	90.91
St. Paul Surigao University Hospital	79	89.77
Surigao Medical Center	78	88.64
Surigao Del Norte Provincial Hospital	50	56.82
Surigao City	37	42.05
Tagana-an	26	29.55
San Francisco	13	14.77
Malimono	34	55.42
Tubod	32	36.36
Sison	40	45.46
Mainit	20	22.73

The table shows the completeness of the various ambulance services' supplies, equipment, and medication in Surigao Del Norte. The DOH's requirement for an appropriately prepared ambulance lists a total of 88 items.

With a score of 90.91 percent, Caraga Regional Hospital has the most supplies available. With an 89.77 percent rating, it is followed by St. Paul Surigao University Hospital. Hospitals, it has been noticed, have more supplies on hand than municipal emergency response agencies. Because hospitals are required to handle a variety of medical situations, they have better facilities. As a result, their ambulances are usually stocked with the necessary supplies, equipment, and drugs.

Several municipal officials stated during the data collection process that their ambulances are only used for scoop-and-run. Instead of ambulance service provider, the right term is Patient Transport Vehicle. The participants said that their ability to provide high-quality care to patients is hampered due to a shortage of resources.

Level of Performance

Table 4. The Level of Performance of the Participants in the Management of Out-of-Hospital Cardiac Arrest

INDICATORS	M	SD	VI	QD
A. SCENE ARRIVAL AND SIZE-UP				
1. I am prepared to review dispatch information.	3.46	0.60	SA	E
2. I am prepared to use lights and sirens when responding as appropriate per emergency medical dispatch information and local guidelines	3.32	0.71	SA	E
3. I am prepared to utilize Body Substance Isolation, as appropriate.	3.43	0.67	SA	E
4. I am prepared to check scene safety and bystander safety.	3.53	0.65	SA	E
5. I am prepared to perform environmental hazards assessment.	3.73	0.68	SA	E
6. I am prepared to determine the number of patients in the area.	3.49	0.61	SA	E
AVERAGE	3.43	0.55	SA	E
B. ASSESSMENT PRIORITIES				
7. I am prepared to determine unresponsiveness, absence of breathing and pulselessness.	3.58	0.55	SA	E

	8. I am prepared to determine patient's hemodynamic stability, symptoms, level of consciousness, Airway, Breathing, Circulation (ABCs), vital signs.	3.50	0.58	SA	E
	9. I am prepared to apply the cardiac monitor and obtain a 12-lead ECG tracing as soon as possible when clinically appropriate.	2.75	1.05	A	P
	10. I am prepared to obtain a thorough assessment related to the event.	3.46	0.61	SA	E
	11. I am prepared to obtain a complete medical history.	3.41	0.67	SA	E
	AVERAGE	3.34	0.56	SA	E
C.	AMBULANCE STRETCHER OPERATION				
	12. I am prepared to move a patient on the ambulance stretcher, adjust the height of the ambulance stretcher from the "load position" to a safe position for travel.	3.55	0.56	SA	E
	13. I am prepared to properly secure all patients using the required straps, including the over-the-shoulder harness, hip and leg restraining straps.	3.56	0.52	SA	E
	AVERAGE	3.56	0.53	SA	E
D.	TREATMENT (HIGH QUALITY CPR AND AED)				
	14. I am prepared to put patient in a supine position before starting CPR.	3.56	0.67	SA	E
	15. I am prepared to start CPR if the patient has no pulse and labored to no breathing.	3.53	0.68	SA	E
	16. I am prepared to give continuous CPR at the rate of 200-240 pushes for 2 minutes without giving ventilation.	3.29	0.81	SA	E
	17. I am prepared to use my body weight to help administer compressions that are at least 2 inches deep in adult patients.	3.45	0.68	SA	E
	18. I am prepared to use Automated External Defibrillator (AED) as soon as possible with minimal interruption of chest compressions.	2.65	1.04	A	P
	19. I am prepared to continue 2 minute cycles of uninterrupted chest compressions followed by AED analysis and shock for 4 cycles (8 minutes).	2.70	1.02	A	P
	20. I am prepared to aid oxygenation by oropharyngeal or nasopharyngeal airway.	3.20	0.86	A	P
	21. I am prepared to give Bag Valve Mask (BVM) ventilation during recoil and without interrupting compressions.	3.42	0.68	SA	E
	22. I am prepared to continue 2 minute cycles of uninterrupted chest compressions, after 4 cycles (8 minutes).	3.36	0.69	SA	E
	AVERAGE	3.24	0.62	A	P
E.	MEDICATION USE AND STORAGE				
	23. I am prepared to check availability of the necessary medications for an adequately equipped ambulance.	3.25	0.89	SA	E
	24. I am prepared to securely maintain and store all medications and fluids at the appropriate temperatures as designated by manufacturer's recommendations.	3.23	0.86	A	P
	AVERAGE	3.24	0.86	A	P
F.	EXCEPTION PROTOCOLS				
	25. I am prepared to withhold CPR if the scene becomes unsafe.	3.57	0.62	SA	E
	26. I am prepared to stop CPR if the patient has positive breathing and positive pulse.	3.60	0.61	SA	E

	27. I am prepared to withhold CPR if I am too exhausted to continue.	3.46	0.69	SA	E
	28. I am prepared to withhold CPR if another trained responder or ERS personnel will take over.	3.60	0.63	SA	E
	29. I am prepared to stop CPR if it has been over 30 minutes giving CPR.	3.48	0.68	SA	E
	AVERAGE	3.54	0.59	SA	E
G.	TRANSPORT DECISION				
	30. I am prepared to transport to the nearest appropriate treatment facility.	3.55	0.56	SA	E
	31. I am prepared to notify receiving facility as early as possible.	3.50	0.61	SA	E
	32. I am prepared to justify the use of lights and sirens for immediate medical intervention that is beyond the capabilities of the ambulance crew using available supplies and equipment.	3.43	0.61	SA	E
	AVERAGE	3.49	0.54	SA	E

Table 4 demonstrates that the ambulance workers performed admirably in the management of out-of-hospital cardiac arrest in five of the seven variables. The highest mean for Indicator E, Ambulance Stretcher Operation, is 3.56 (SD=0.53), indicating that individuals perform excellently in properly loading patients for transport. As a result, it emphasizes the importance of ambulances as a patient transportation system. It is essential in saving lives in emergency situations.

Treatment (High Quality CPR and Defibrillation) and Medication Use and Storage, respectively, have the lowest mean of 3.24 (SD=0.62, 0.86) which is a tie in the rating. In carrying out these two metrics, ambulance staff have just a Proficient performance. This means that the participants have a more complete awareness of the treatment and medication use, which helps them make better decisions. However, they lack the skills that come from experience and a deeper understanding of the above-mentioned indicators, which are required to become expert staff.

Item 18, 19, and 20 resulted in Proficient performance under indication E (Treatment). Item 18 discusses how to use an AED while keeping chest compressions as little as possible. Item 19 involves doing a 4-cycle cycle of continuous 2-minute chest compressions followed by AED analysis and shock (8 minutes). Item 20 is about using the oropharyngeal or nasopharyngeal airway to help with oxygenation. Personnel are rarely exposed to these treatments because they are not carried out on a daily basis. Defibrillation also necessitates extensive training, which includes ECG reading and other procedures. As a result, they are less prepared and confident in their ability to carry out the aforementioned interventions. The same is true for item 9 under indication B (Assessment Priorities), which is about connecting a cardiac monitor to the patient and obtaining a 12-lead ECG trace.

For two reasons, participants are less likely to complete indication E (Medication Use and Storage). To begin with, they are not responsible for the use of medications. They are not allowed to prescribe or administer drugs. Second, medications aren't always easily available in their ambulance, which is why they don't use them.

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Table 5. Summary of average mean of performance of ambulance personnel in OHCA management

Parameters	M	SD	VI	QD
Scene arrival and size up	3.43	0.55	SA	E
Assessment priorities	3.34	0.56	SA	E
Ambulance stretcher operation	3.56	0.53	SA	E
Treatment (High quality CPR and AED)	3.24	0.62	A	P
Medication use and storage	3.24	0.86	A	P
Exception protocols	3.54	0.59	SA	E
Transport decision	3.49	0.54	SA	E

Table 6. Significant Relationship between the Profile of the Participants and the Level of Performance

Profile Variables	Mean	SD	R(X,Y)	P	Decision	Interpretation
Age	2.986	1.179	-0.1499	0.1991	Accept H_0	There is no significant relationship
Highest Educational Attainment	3.546	3.068	-0.1129	0.3347	Accept H_0	There is no significant relationship
Years of Experience in ERS	2.573	1.055	-0.1363	0.0243	Reject H_0	There is a significant relationship

Profile Variables	SS effect	df effect	MS effect	F	p	Eta	Eta squared	Decision	Interpretation
Sex	0.335	1	0.335	1.854	0.185	0.263	0.069	Accept H_0	There is no significant relationship
Civil Status	0.023	1	0.023	0.119	0.733	0.069	0.005	Accept H_0	There is no significant relationship

When individuals are categorized by age and greatest educational attainment, the p-values (0.1991 and 0.3347, respectively) are higher above the 0.05 significant level, as indicated in the table. This suggests that there is no link between participants' age and highest educational achievement and their level of performance.

When years of experience in ERS are aggregated, the p-value is 0.0243, which is less than the 0.05 significant level. This suggests that the degree of performance and the number of years of experience have a strong correlation. Acquiring skills via experience, according to Patricia Benner's Novice to Expert Nursing Theory, is a prerequisite for becoming an expert nurse. Work experience allows people to practice and improve their skills. As a result, exposure to cardiac arrest cases is one of the elements that improves their performance, resulting in a better patient outcome.

An Eta result of 0.263 indicates a weak relationship between the two variables when grouped by sex. The null hypothesis is so accepted. This suggests that there is no relationship between the participants' sex and their level of performance.

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Similarly, an Eta result of 0.069 indicates a minimal relationship between the two variables in terms of their civil status. The null hypothesis is so accepted. This suggests that there is no relationship between the participants' civil status and their level of performance.

Findings

Gathering, tabulating, and analyzing data. The study's findings are summarized below:

1. The majority of the 75 participants in the study were between the ages of 30 and 34, accounting for 42.47 percent of the total number of participants. In terms of gender, the majority of participants are female, accounting for 64% of the overall number of participants, while males account for 36% of the total number of participants. In terms of civil status, the majority of participants are married, accounting for 56 percent of the total number of participants, while those who are single account for 44 percent. The bulk of the participants, 76 percent of the total number of participants, have a Bachelor of Science in Nursing as their greatest educational attainment. Furthermore, the majority of the participants had 1-3 years of experience in ERS, accounting for 44.7 percent of the total number of participants. Finally, when it comes to ERS training, all participants are trained in basic life support, followed by basic first aid, which 86.67 percent of the participants have completed.
2. According to the survey, Caraga Regional Hospital has the most supplies available, with a percentage of 90.91 percent based on the DOH standard. With an 89.77 percent rating, it is followed by St. Paul Surigao University Hospital.
3. The study found that five out of seven factors indicate that ambulance staff perform excellently in the management of out-of-hospital cardiac arrests. The Ambulance Stretcher Operation had the highest mean of 3.56 (SD=0.53), indicating that the participants are well-prepared and confident when managing stretchers. Treatment (High-Quality CPR and Defibrillation) and Medication Use and Storage tied for the lowest mean of 3.24 (SD=0.62 and 0.86, respectively), indicating that the participants' performance in these two areas is only Proficient.
4. Items 18, 19, and 20 performed proficiently under treatment (high-quality CPR and Defibrillation). Item 18 discusses how to use an AED while keeping chest compressions as little as possible. Item 19 involves doing a 4-cycle cycle of continuous 2-minute chest compressions followed by AED analysis and shock (8 minutes). Item 20 is about using the oropharyngeal or nasopharyngeal airway to aid with oxygenation. Since these interventions are not carried out on a daily basis, personnel are less prepared and confident in their ability to carry them out.

5. Similarly, item 9 under Assessment Priorities yielded a Proficient result. It involves attaching a cardiac monitor to the patient and obtaining a 12-lead ECG tracing. This is not a common intervention, which makes it difficult for the ambulance personnel to do.
6. With a p-value of 0.0243, there is a significant relationship between level of performance and years of experience in ERS. According to Patricia Benner's nursing theory, performance improves by practicing and strengthening acquired skills through work exposure. This eventually results in a positive patient outcome. The other profile characteristics do not have a significant relationship with performance level.

Conclusion

The following conclusion has been drawn based on the study's findings:

1. The majority of the participants are between the ages of 30 and 40, are married, and have completed a bachelor's degree in nursing. Has 1-3 years of experience working in the emergency response field. All participants are trained in Basic Life Support (BLS) and Basic First Aid (BFA).

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2. Despite receiving an excellent rating on the totality of the performance such as treatment and medication use should not be disregarded. To improve OHCA management and, ultimately, patient survival, quality and continuing CPR and AED training is required. Ambulance personnel's years of experience in EMS have a substantial impact on their level of performance. The knowledge and skills developed through experience play a significant role in the personnel's competency.

Recommendations

The findings of the study could be used to help identify strategies to improve ambulance personnel's OHCA management. The following are the recommendations offered:

1. Personnel in the Ambulance Service

a. More training sessions. The trainings should be conducted on a yearly basis to keep the workers updated.

2. Ambulance Facilities

a. Comprehensive and high-quality ambulance facilities. Ambulance providers, particularly those from towns, should modernize their facilities and meet the Department of Health's standard for a well -equipped ambulance.

3. Implementation of the Proposed Continuing Ambulance Personnel Competency Training. Treatment (high-quality CPR and AED) as well as medication administration and storage are all part of the proposed personnel competency program.

4. For future researchers, widen the scope and replication of the study.

Ambulance Personnel Continuing Competency Training Program on Basic Life Support and Use of AED

II. Background/Rationale

From the study "Demographics as Correlates to Performance among Ambulance Personnel Providing Out-of-Hospital Cardiac Arrest (OHCA) Management: Basis for a Proposed Personnel Competency Program," Results showed ; ambulance personnel have Exemplary performance on 5 out of 7 indicators in managing out-of-hospital cardiac arrest, with the Ambulance Stretcher Operation having the highest mean of 3.56 (SD=0.53). The lowest mean was 3.24 (SD=0.62 and 0.86, respectively) for Treatment (High Quality CPR and Defibrillation) and Medication Use and Storage. This indicates that the individuals were capable of completing these two tasks. The goal of this training program is to improve ambulance personnel's skills in administering BLS and AED to OHCA patients in the hope of enhancing their chances of survival.

III. Scope

a. Audience

i. All ambulance and emergency transport service employees in Surigaodel Norte will be invited to participate in this continuing skills and competency enhancement program on Basic Life Support (BLS) and the use of an Automated Emergency Defibrillator (AED).

b. Objectives

To prepare ambulance and emergency transport service employees with the knowledge, skills, and attitude necessary to properly administer basic life support to out-of-hospital-cardiac-arrest (OHCA) patients, with or without the use of an Automated External Defibrillator (AED).

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c. Competencies to be acquired
The participants should be able to do the following after the training:
i. Describe the principles of emergency care.
ii. Identify the four links of survival OHCA patients.
iii. Demonstrate how to provide rescue breathing, chess compression, and usage of AED to OHCA patients in times of CoViD-19 Pandemic.
a. **Duration**
The two-day enhanced training session will take place.
b. Study Materials

The training committee is responsible for preparing the following supplies, equipment, and provisions.

1. BLS and AED information sheets in writing
2. A video showcasing the most recent BLS and AED application for the CoViD-19 Pandemic
3. Manikin of Laerdal Resuscitation
4. To disinfect the manikin, use 70% alcohol.
5. cleaning the manikin with 4x4 gauze
6. A white board, white board markers, and a white board eraser
7. Snacks and hydration

ii. The participants must prepare the following.
1. Participants are required to bring and utilize their own laptops, tablets, or other devices to see the video.

c. Mode of delivery and duration
i. BLS and AED information in writing. Participants will get a written information sheet outlining the most up-to-date BLS and AED algorithms for use in the event of a CoViD-19 pandemic. During meetings, participants will be requested to read and go over the information sheet by themselves or with their colleagues. The task will be handed to the participants for six (6) hours to finish.

ii. Video-assisted instruction In the event of a CoViD-19 pandemic, participants will be presented a video recording outlining the most recent BLS and AED protocol. Participants are free to watch and rewatch the video as many times as they want. The task will be handed to the participants for six (6) hours to finish.

iii. BLS and AED teacher simulation training. Under the supervision of the BLS and AED instructor, the participants will receive hands-on practice with BLS and AED application using the Laerdal Resuscitation Manikin. The task will be handed to the participants for eight (8) hours to finish.

d. Course Outline
i. Introduction to BLS and AED usage in CoViD-19 Pandemic
ii. Principles of Emergency Care
iii. Four links of survival
iv. CoViD-19 transmission cycle and prevention
v. Guidance for performing breathing assessment and rescue breathing
v. Compression recommendations
vii. AED Application and Use Instructions
viii. Cleaning and disinfection advice

e. **Assessments**
i. To measure their level of comprehension and competency in the delivery of BLS and AED, the participants will take a formative pre-assessment test as well as a summative post-assessment test.

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BIBLIOGRAPHY

[1] (2012). Retrieved March 9, 2019, from Department of Health: https://www.doh.gov.ph/sites/default/files/publications/PHS2012.pdf

[2] Abella, J. M. (2010, March 19). *Only 3 out of 10 job-hunting grads may find work—IBON*. Retrieved from GMA News Online: https://www.gmanetwork.com/news/news/nation/186506/only-3-out-of-10-job-hunting-grads-may-find-work-ibon/story/

[3] Altun, I. (2003). The perceived problem solving ability and values of student nurses and midwives. *Nurse Education Today*, 578.

[4] Anderson, L. A. (2016, August 23). Retrieved from https://www.drugs.com/support/editorial_policy.html#l-anderson-pharmd

[5] Cardona, R., & Andres, E. (2014). The Employability of Mathematics Education Graduates (2008-2013) of a Teacher Education Institution (TEI) in the Philippines. *Academic Journal Article*.

[6] Cashin-Garbutt, A. (2017). *News Medical Life Sciences*. Retrieved from https://www.news-medical.net/health/Heart-attack-and-cardiac-arrest-difference.aspx#

[7] Cena, E. (2011, July 3). *Business inquirer*. Retrieved from Business inquirer: https://business.inquirer.net/5804/pharmacy-offers-filipinos-affordable-antibiotic-kits?utm_expid=.XqNwTug2W6nwDVUSgFJXed.1

[8] Chaudhury, R. L. (2016, March 2). *PubMed*. Retrieved from NCBI: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4775002/

[9] Chin, B., Murphy, M., Janicki-Deverts, D., & Cohen, S. (2017, 2 13). *Science News*. Retrieved 3 21, 2020, from Sceience Daily: https://www.sciencedaily.com/releases/2017/02/170213131232.htm

[10] *Department of Health* . (2016, June 14). Retrieved May 5, 2020, from https://www.doh.gov.ph/morbidity

[11] DOH. (2018, March 4). *DOH*. Retrieved from DOH: https://www.doh.gov.ph/node/1058

[12] Dray, B., Burke, L., Hurst, H., Ferguson, A., & Maran, D. (2011). Enhancing the Employability of the Newly Qualified Nurses: A Pilot Study. *Journal of Further and Higher Education*.

[13] Dukes, K., & Girotra, S. (2019). Are Lay Rescuers Adequately Prepared for CPR and its Aftermath? *PubMed Central*.

[14] Dumaraos, G. (2018). A CPR -Ready Philippines by 2021. *Healthy and Lifestyle*, 14-15.

[15] Dyar, O. J. (2016, February 24). *TFO*. Retrieved from Tandfonline: https://www.tandfonline.com/doi/full/10.1586/14787210.2016.1151353

[16] Dyson, K. (2017, April 17). Retrieved March 2019, 7, from http://au-east.erc.monash.edu.au/fpfiles/8221271/20170417DysonThesis.pdf

[17] *Environment Health and Safety*. (2020, January). Retrieved from https://www.sgs.ph: https://www.sgs.ph/en/environment-health-and-safety/emergency-response-services-ers

[18] Frõhlich et al. (2013). Out-of-Hospital Cardiac Arrest –Optimal Management. *USA National Center for Biotechnology Information: National Library of Medicine*.

[19] Gesta, M. I. (2016, January-December). *Annual Report*. Retrieved March 9, 2019, from City Government of Surigao: http://surigaocity.gov.ph/sites/default/files/documents/milestones/2016%20Annual%20Report.pdf?fbclid=IwAR29U1MpdmLAF5hHQI7rCRafieCJtlEBHr4DFGWiEjurJ30oFQA2U8xvBsk

[20] Haque, M. (2017, August 10). *researchgate*. Retrieved from researchgate: https://www.researchgate.net/publication/318772954_Antimicrobial_Use_Prescribing_And_Resistance_In_Selected_Ten_Selected_Developing_Countries_A_Brief_Overview?_esc=publicationCoverPdf&el=1_x_2&enrichId=rgreq-698ffeabe205887b7a295a75616d860d-XXX&enrichSour

[21] Hermoso, C. (2018, June 22). *Manila Bulletin*. Retrieved from news.mb.com.ph.

[22] Hertel, J., Schutz, A., De Paulo, B. M., Morris, W., & Stucke, T. S. (2007). She's single, so what? How are singles perceived compared with people who are married? 139-158.

[23] Joanne, S., Lela, C., Mathew, J., & Jacqueline, M. (2017, November 1). *Health force*. Retrieved February 17, 2019, from healthforce.uscf.edu: https://healthforce.ucsf.edu/publications/california-board-registered-nursing-2016-survey-registered-nurses

[24] Kaur, A. (2018, May 10). *PMC*. Retrieved from NBCI: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6044109/?fbclid=IwAR1HsIvFOjmbH7vCa9tW7EvkSc6sXOaSYMAnxsa6dtzVPHOK66pyl4UeDJM

[25] Kronick. (2015, November). *CPR and First AId Emergency Cardiovascular Care*. Retrieved from https://eccguidelines.heart.org/circulation/cpr-ecc-guidelines/

[26] L. H. Soo et al. (1998). Influence of ambulance crew's length of experience. *European Heart Journal*, 536-539.

[27] Lalican, N. M. (2012). Tracer Study of Agriculture Students.

[28] Larson, E. (2007, April 21). *Annual Review of Public Health*. Retrieved from Annual reviews: https://www.annualreviews.org/doi/10.1146/annurev.publhealth.28.021406.144020

[29] Lorber, M., & Savic, B. S. (2012). Job satisfaction of nurses and identifying factors of job satisfaction in Slovenian Hospitals. *Croatian Medical Journal*.

[30] M. Daya et al. (2015, February 9). *US National Library of Medicine*. Retrieved March 10, 2019, from National Center for Biotechnology Information: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4433591/

[31] Mehra, R. (2017, December). *National Institute of Health*. Retrieved from US National Library of Medicine: http://ncbi.nlm.nih.gov/pubmed/17993308

[32] Mendoza, G. (2008). An analysis of theperformance of selected nursing graduates at the 2007 nurse licensure examination.

[33] Napallacan, J. (2012, February 17). *Cebu Daily News*. Retrieved from Cebu Daily News: https://cebudailynews.wordpress.com/2012/02/17/more-nurses-few-jobs/

[34] Nessim, N. (2019, January 15). *solvhealth*. Retrieved from solvhealth: https://www.solvhealth.com/blog/antibiotics-everything-you-need-to-know

[35] NHS. (2014, June 5). *NHS*. Retrieved from NHS: https://www.nhs.uk/conditions/antibiotics/

[36] Nolan, J., Soar, J., & Eikeland, H. (2006). The Chain of Survival. *Resuscitation: Official Journal of the European Resuscitation Council*, 270–271.

[37] Orillaza, J. M. (2014, May 1). *Labor mismatch or what ails the PHL jobs market*. Retrieved from GMA NEWS ONLINE: https://www.gmanetwork.com/news/money/companies/359201/labor-mismatch-or-what-ails-the-phl-jobs-market/story/

[38] Out-of-hospital cardiac arrest: a unique medical emergency. (2018). *The Lancet*, 911.

[39] Panaligan, M. (2018). *psmid*. Retrieved from psmid: https://www.psmid.org.ph/index.php/news-media/news-online/169-antimicrobial-resistance-as-an-emerging-problem

[40] Pareja, D. E. (2018, September 27). *News mb*. Retrieved from News mb: https://news.mb.com.ph/2018/09/27/dengue-cases-continue-to-rise-in-caraga-region/

[41] Patil, Halperin, & Becker. (2018). Cardiac Arrest: Resuscitation and Reperfusion. *PubMed Central*.

[42] Pharmaceutical journal. (2018, November 14). *Pharmaceutical journal*. Retrieved from Pharmaceutical journal: https://www.pharmaceutical-journal.com/20205732.article

[43] Pollack et al. (2019). Impact of bystander automated external defibrillator use on survival and functional outcomes in shockable observed public cardiac arrests. *PudMed Central*, 2104-2113.

[44] Rashid, S., & Rashid, U. (2012). Work Motivation Differences between Public and Private Sector.

[45] Regalario, B. R. (2012). *pidsphil*. Retrieved from pidsphil: http://www.pidsphil.org/pdf/Journal_01012012/jo42_ja07.pdf

[46] Regmi, P. (2009). Tracer Study of AIT graduates 2003-2008.

[47] Research Gate. (2007, February 7). *Research Gate*. Retrieved from Research Gate: https://www.researchgate.net/publication/20837369_Purchase_of_antibiotics_without_prescription_in_Manila_The_Philippines_Inappropriate_choices_and_doses

DOI: 10.51386/25815946/ijsms-v5i1p114

Volume: 5 Issue: 1 January to February 2022 https://www.ijsmsjournal.org

[48] Research Institute for Tropical Medicine. (2018, February 28). *Research Institute for Tropical Medicine*. Retrieved from Research Institute for Tropical Medicine: http://ritm.gov.ph/national-antibiotic-guidelines-2017/?fbclid=IwAR2FttvArzo0ZfPGlMLNCiECHupqA5uE6oqnnRdIpaCyxMmQ0Hd_EFCu5Dw

[49] Robertson, K. E., & McDaniel, A. (1995). Interdisciplinary Professional Education: A Collaborative Clinical Teaching Project. *American Journal of Pharmaceutical Education*.

[50] Rodgers, L. M. (1997, July 21). Retrieved March 10, 2019, from https://watermark.silverchair.com/48-1-7.pdf?token=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAAkAwggI8BgkqhkiG9w0BBwagggItMIICKQIBADCCAiIGCSqGSIb3DQEHATAeBglghkgBZQMEAS4wEQQMS5TIKIrCd-0Im6__AgEQgIIB815Kn_BTcuSGXAfcA-HWUZ7tLYgWwMsJxdMlb5T0VUId7BjP

[51] Roope, D. L. (2019, April 5). *Phys.Org*. Retrieved from Phys.Org: https://phys.org/news/2019-04-economics-tackle-global-antimicrobial-resistance.html

[52] Satterwhite, R. (2013, April 2). Retrieved March 10, 2019, from Ada County Paramedics: https://adacountyparamedics.org/blog/2013/04/02/being-a-female-paramedic/

[53] Schomburg, H. (2003). *Handbook of Tracer Studies.*

[54] Shafiq, N. (2018, May 10). *PMC*. Retrieved from NBCI: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6044109/?fbclid=IwAR1HsIvFOjmbH7vCa9tW7EvkSc6sXOaSYMAnxsa6dtzVPHOK66pyl4UeDJM#bibr12-2049936118773216

[55] Shier, W. C. (2018, December 27). *Medical Dictionary A-Z List*. Retrieved from MedicineNet: https://www.medicinenet.com/script/main/art.asp?articlekey=10705

[56] Sison, P. (2005). *Personnel Management in the 21st Century.* REX Book Store Inc.

[57] Smith, J., McKnight, A., & Naylor, R. (2000). Graduate Employability: Policy and Performance in Higher Education in the UK. *The Economic Journal*, 382-411.

[58] *Strategies to Improve Cardiac Arrest Survival: A Time to Act*. (2015). Retrieved from National Academic Press: https://www.nap.edu/read/21723/chapter/6

[59] Teichler, U. (1981). *Higher Education and the World of Work.* Rotterdam/Taipei: Sense Publishers.

[60] Tesser, M. (2018, May 3). *Training Of Emergency Medical Services In The Philippines*. Retrieved April 16, 2020, from Emergency Live: https://www.emergency-live.com/ambulance/training-emergency-medical-services-philippines/

[61] Tomey, A. M. (2009). *Guide to Nursing Management and Leadership.* Singapore: Elsevier Pte Ltd. .

[62] Tomlinson, M. (2012). Graduate Employability: A Review of Conceptual and Emperical Themes. *Higher Education Policy*.

[63] Uchmanowicz, I., Bartkiewicz, W., Sowizdraniuk, J., & Rosińczuk , J. (2015). Factors Affecting the Occurrence of Out-of-Hospital Sudden Cardiac Arrest. *PudMed Central*.

[64] Ventola, C. L. (2015, April). *US National Library of Medicine Nationl Instittute of Health* . Retrieved May 5, 2020, from https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4378521/

[65] Wharton, A. (2013). *Gender and Work.* Oxford University Press.

[66] *What is EMS?* (2020). Retrieved from ems.gov: https://www.ems.gov/whatisems.html

[67] WHO. (2011). *WHO.* Retrieved from file:///I:/How-to-investigate-antimicrobial-use.pdf

[68] WHO. (2018, February 5). *World Health Organization*. Retrieved from World Health Organization: https://www.who.int/en/news-room/fact-sheets/detail/antibiotic-resistance

[69] Who.int. (2015, March 10). *World Health Organization*. Retrieved from World Health Organization: https://www.who.int/en/news-room/fact-sheets/detail/antimicrobial-resistance

[70] Williams, M. (2015, July 27). *CRP Certified Inidigo Medical Training*. Retrieved February 17, 2019, from www.cprvertified.com: https://www.cprcertified.com/blog/the-benefits-of-bls-training?fbclid=IwAR32YM-ovCC6szY12gyXb6IKUy84_xYfmEwOO74u5UEzSlcmdLNNgn6llas

DOI: 10.51386/25815946/ijsms-v5i1p114

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[71] *World Health Organization* . (2018). Retrieved May 5, 2020, from https://www.who.int/gho/mortality_burden_disease/causes_death/top_10/en/

[72] *World Health Organization* . (2018, February 5). Retrieved May 5, 2020, from https://www.who.int/news-room/fact-sheets/detail/antibiotic-resistance

[73] Zimmerman, J. (2012, January 31). Are College students Learning?

[74] American Heart Association.(2020, April). BLS Healthcare Provider Adult Cardiac Arrest Algorithm for Suspected of Confirmed COVID-19 Patients. https://cpr.heart.org/-/media/cpr-files/resources/covid-19-resources-for-cpr-training/english/algorithmbls_adult_cacovid_200406.pdf?la=en

[75] Edelson, D., Sasson, C & Chan P. (2020, April 9).Interim Guidance for Basic and Advanced Life Support in Adults, Children, and Neonates With Suspected or Confirmed COVID-19. Circulation. 2020;141:e933–e943. https://doi.org/10.1161/CIRCULATIONAHA.120.047463

[76] The American Red Cross Training Services.(n.d.). First Aid/CPR/AED Care During COVID-19. https://www.redcross.org/take-a-class/coronavirus-information/first-aid-cpr-aed-care-during-covid-19